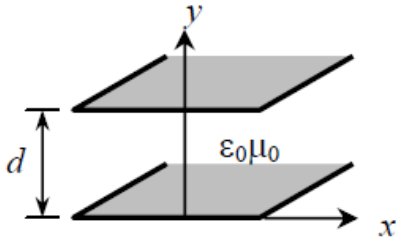


第五章习题参考答案

5.2



$$(1) \nabla \cdot \mathbf{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = 0$$

$$\begin{aligned} \nabla \times \mathbf{E} &= \mathbf{x}_0 \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} \right) + \mathbf{y}_0 \left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} \right) + \mathbf{z}_0 \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right) \\ &= -\mathbf{y}_0 jkA \sin\left(\frac{\pi}{d}y\right) e^{j(\omega t - kz)} - \mathbf{z}_0 \frac{\pi}{d} A \cos\left(\frac{\pi}{d}y\right) e^{j(\omega t - kz)} \end{aligned}$$

(2) $\nabla \times \mathbf{E} \neq 0$, 是有旋场, 不能用标量函数的负梯度表示

$$(3) \mathbf{H} = -\frac{1}{j\omega\mu_0} \nabla \times \mathbf{E} = \mathbf{y}_0 \frac{kA}{\omega\mu_0} \sin\left(\frac{\pi}{d}y\right) e^{j(\omega t - kz)} + \mathbf{z}_0 \frac{\pi}{d} \frac{1}{j\omega\mu_0} A \cos\left(\frac{\pi}{d}y\right) e^{j(\omega t - kz)}$$

$$(4) \mathbf{J}_s = \mathbf{n} \times \mathbf{H}$$

$$\begin{aligned} \mathbf{J}_s|_{y=0} &= \mathbf{y}_0 \times \left[\mathbf{y}_0 \frac{kA}{\omega\mu_0} \sin\left(\frac{\pi}{d}y\right) e^{j(\omega t - kz)} + \mathbf{z}_0 \frac{\pi}{d} \frac{1}{j\omega\mu_0} A \cos\left(\frac{\pi}{d}y\right) e^{j(\omega t - kz)} \right] \\ &= \mathbf{x}_0 \frac{\pi}{d} \frac{1}{j\omega\mu_0} A e^{j(\omega t - kz)} \end{aligned}$$

$$\text{同理 } \mathbf{J}_s|_{y=d} = -\mathbf{x}_0 \frac{\pi}{d} \frac{1}{j\omega\mu_0} A e^{j(\omega t - kz)}$$

$$\rho_s = \mathbf{D} \cdot \mathbf{n}$$

$$\rho_s|_{y=0} = \mathbf{D} \cdot \mathbf{n} = 0$$

$$\rho_s|_{y=d} = \mathbf{D} \cdot \mathbf{n} = 0$$

5.3

(1) 入射波是右手圆极化。

对于反射波, 为满足导体表面边界条件, E_x^r, E_y^r 与 E_x^i, E_y^i 都有 180° 相移, 且波传播方向相反, 所以是左手圆极化。

$$(2) \mathbf{H} = -\frac{1}{j\omega\mu} \nabla \times \mathbf{E} = -\frac{1}{j\omega\mu} \begin{vmatrix} \mathbf{x}_0 & \mathbf{y}_0 & \mathbf{z}_0 \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_0 e^{-jkz} & -jE_0 e^{-jkz} & 0 \end{vmatrix} = (j\mathbf{x}_0 + \mathbf{y}_0) \frac{k}{\omega\mu_0} E_0 e^{-jkz}$$

(3) 此入射波可看成是两个平面波的叠加。 $\mathbf{E}_1 = \mathbf{x}_0 E_0 e^{-jkz}$, $\mathbf{E}_2 = -j\mathbf{y}_0 E_0 e^{-jkz}$, 在这个坐标系下两个均为 TEM 波,

对平面波 1, 在 $z \leq 0$ 区域合成电场强度 $E_x(z) = E_0(e^{-jkz} - e^{jkz}) = -2jE_0 \sin kz$

对平面波 2, 在 $z \leq 0$ 区域合成电场强度 $E_y(z) = -jE_0(e^{-jkz} - e^{jkz}) = -2E_0 \sin kz$

所以 $z \leq 0$ 区域合成电场强度的瞬时值 $E_x(z) = 2\mathbf{x}_0 E_0 \sin kz \sin \omega t - 2\mathbf{y}_0 E_0 \sin kz \cos \omega t$

5.5

将该圆极化波分解为 TE、TM 波，如果布儒斯特角 $\theta_b = 60^\circ$ ，则反射波只有 TE 波，由 $\theta_b = 60^\circ$ ，得到

$$\theta_b = \arctan \sqrt{\frac{\varepsilon_{r2}}{\varepsilon_{r1}}} = \frac{\pi}{3}, \quad \varepsilon_{r2} = \sqrt{3}$$

5.7

由相位匹配，可以得到：

$$k_1 \sin \theta_1 = k_2 \sin \theta_2 = k_3 \sin \theta_3, \quad \text{所以 } \theta_3 = \arcsin \left(\frac{k_1}{k_3} \sin \theta_1 \right)$$

5.8

图略

$$\text{水中, } k_{wx} = \omega \sqrt{1.7 \varepsilon_0 \mu_0} \cos 30^\circ = \omega \frac{\sqrt{5.1}}{2} \sqrt{\varepsilon_0 \mu_0}, \quad k_{wyz} = \omega \sqrt{1.7 \varepsilon_0 \mu_0} \sin 30^\circ = \omega \frac{\sqrt{1.7}}{2} \sqrt{\varepsilon_0 \mu_0}$$

$$\text{因为, } k_{wyz} = k_{ayz}, \quad \text{所以空气中, } k_{ax} = \sqrt{k_a^2 - k_{ayz}^2} = \omega \sqrt{\varepsilon_0 \mu_0} \left(1 - \frac{1.7}{4} \right)^{1/2} = \omega \frac{\sqrt{2.3}}{2} \sqrt{\varepsilon_0 \mu_0}$$

$$\text{对于 TE 波, } Y_a = \frac{k_{ax}}{\omega \mu_0} = \frac{\sqrt{2.3}}{2} \sqrt{\frac{\varepsilon_0}{\mu_0}}, \quad Y_w = \frac{k_{wx}}{\omega \mu_0} = \frac{\sqrt{5.1}}{2} \sqrt{\frac{\varepsilon_0}{\mu_0}}$$

$$\text{反射系数: } \Gamma_{TE} = \frac{Y_w - Y_a}{Y_w + Y_a} = \frac{\sqrt{5.1} - \sqrt{2.3}}{\sqrt{5.1} + \sqrt{2.3}} = 0.196, \quad \text{透射系数: } T_{TE} = 1 + \Gamma_{TE} = 1.196$$

$$\text{对于 TM 波, } Y_a = \frac{\omega \varepsilon_0}{k_{ax}} = \frac{2}{\sqrt{2.3}} \sqrt{\frac{\varepsilon_0}{\mu_0}}, \quad Y_w = \frac{\omega \varepsilon}{k_{wx}} = \frac{2\sqrt{5.1}}{3} \sqrt{\frac{\varepsilon_0}{\mu_0}}$$

$$\text{反射系数: } \Gamma_{TM} = \frac{Y_w - Y_a}{Y_w + Y_a} = \frac{\frac{2\sqrt{5.1}}{3} - \frac{2}{\sqrt{2.3}}}{\frac{2\sqrt{5.1}}{3} + \frac{2}{\sqrt{2.3}}} = \frac{\sqrt{11.73} - 3}{\sqrt{11.73} + 3} = 0.066,$$

$$\text{透射系数: } T_{TM} = 1 - \Gamma_{TM} = 0.934$$

5.9

$$1) \sqrt{\varepsilon_{r1}} \sin 45^\circ = \sqrt{\varepsilon_{r2}} \sin 30^\circ, \quad \varepsilon_{r2} = 2$$

2) 由图所示，该平面波为 TM 波，

$$\Gamma = \frac{\varepsilon_{r1} k_{z2} - \varepsilon_{r2} k_{z1}}{\varepsilon_{r1} k_{z2} + \varepsilon_{r2} k_{z1}} = \frac{\frac{\sqrt{3}}{2} \sqrt{2} k_0 - 2 \frac{\sqrt{2}}{2} k_0}{\frac{\sqrt{3}}{2} \sqrt{2} k_0 + 2 \frac{\sqrt{2}}{2} k_0} = -0.0718$$

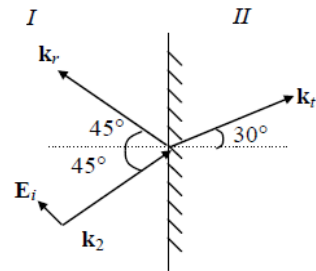


图 P5.9

5.10

对于 TE 模

$$\Gamma_{TE} = \frac{Z_2 - Z_1}{Z_2 + Z_1} = \frac{\mu_2 k_{z1} - \mu_1 k_{z2}}{\mu_2 k_{z1} + \mu_1 k_{z2}}$$

$$\text{反射为零时, } \mu_2 k_{z1} - \mu_1 k_{z2} = 0$$

即 $\mu_2 k_1 \cos \theta_B = \mu_1 k_2 \cos \theta_2$

由相位匹配条件: $k_1 \sin \theta_B = k_2 \sin \theta_2$

$$\cos \theta_2 = \frac{\mu_2 k_1}{\mu_1 k_2} \cos \theta_B, \quad \sin \theta_2 = \sqrt{1 - \cos^2 \theta_2} = \sqrt{1 - \frac{\mu_2^2 k_1^2}{\mu_1^2 k_2^2} \cos^2 \theta_B}$$

$$k_1 \sin \theta_B = k_2 \sin \theta_2 = k_2 \sqrt{1 - \frac{\mu_2^2 k_1^2}{\mu_1^2 k_2^2} \cos^2 \theta_B}$$

$$\frac{k_1}{k_2} \sqrt{1 - \cos^2 \theta_2} = \sqrt{1 - \frac{\mu_2^2 k_1^2}{\mu_1^2 k_2^2} \cos^2 \theta_B}$$

两边平方, 均整理后得到

$$\cos^2 \theta_B = \frac{\mu_1}{\varepsilon_1} \frac{\mu_1 \varepsilon_1 - \mu_2 \varepsilon_2}{\mu_1^2 - \mu_2^2}$$

$$\text{所以 } \theta_B = \arccos \sqrt{\frac{\mu_1}{\varepsilon_1} \frac{\mu_1 \varepsilon_1 - \mu_2 \varepsilon_2}{\mu_1^2 - \mu_2^2}}$$

对于 TM 模

$$\Gamma_{TE} = \frac{Z_2 - Z_1}{Z_2 + Z_1} = \frac{\varepsilon_1 k_{z2} - \varepsilon_2 k_{z1}}{\varepsilon_1 k_{z2} + \varepsilon_2 k_{z1}}$$

反射为零时, $\varepsilon_1 k_{z2} - \varepsilon_2 k_{z1} = 0$

即 $\varepsilon_2 k_1 \cos \theta_B = \varepsilon_1 k_2 \cos \theta_2$

由相位匹配条件: $k_1 \sin \theta_B = k_2 \sin \theta_2$

$$\cos \theta_2 = \frac{\varepsilon_2 k_1}{\varepsilon_1 k_2} \cos \theta_B, \quad \sin \theta_2 = \sqrt{1 - \cos^2 \theta_2} = \sqrt{1 - \frac{\varepsilon_2^2 k_1^2}{\varepsilon_1^2 k_2^2} \cos^2 \theta_B}$$

$$k_1 \sin \theta_B = k_2 \sin \theta_2 = k_2 \sqrt{1 - \frac{\varepsilon_2^2 k_1^2}{\varepsilon_1^2 k_2^2} \cos^2 \theta_B}$$

$$\frac{k_1}{k_2} \sqrt{1 - \cos^2 \theta_2} = \sqrt{1 - \frac{\varepsilon_2^2 k_1^2}{\varepsilon_1^2 k_2^2} \cos^2 \theta_B}$$

两边平方, 均整理后得到

$$\cos^2 \theta_B = \frac{\varepsilon_1}{\mu_1} \frac{\mu_1 \varepsilon_1 - \mu_2 \varepsilon_2}{\varepsilon_1^2 - \varepsilon_2^2}$$

$$\text{所以 } \theta_B = \arccos \sqrt{\frac{\varepsilon_1}{\mu_1} \frac{\mu_1 \varepsilon_1 - \mu_2 \varepsilon_2}{\varepsilon_1^2 - \varepsilon_2^2}}$$

5.11

$$(1) \quad \theta_c = \arcsin \sqrt{\frac{\varepsilon_2}{\varepsilon_1}} = 30^\circ$$

$$(2) \quad \theta = 60^\circ, \quad k_x = k_1 \sin \theta_1 = k_0 \sqrt{4} \sin 60^\circ = \sqrt{3} k_0$$

$$k_{z1} = k_1 \cos 60^\circ = k_0 \sqrt{4} \cos 60^\circ = k_0$$

$$(3) \quad k_{z2} = \sqrt{k_0^2 - k_x^2} = \sqrt{k_0^2 - 3k_0^2} = \sqrt{-2} k_0 = -\sqrt{2} j k_0$$

$$(4) \quad \alpha_2 = \sqrt{2} k_0, \quad \alpha_2 d = 1, \quad d = \frac{1}{\alpha_2} = \frac{1}{\sqrt{2} k_0}$$

$$(5) \quad \Gamma = \frac{k_{z1} - k_{z2}}{k_{z1} + k_{z2}} = \frac{k_0 - j\sqrt{2}k_0}{k_0 + j\sqrt{2}k_0} = e^{-j109.5^\circ}$$

5.12

答：由 5.4.10 式，铜的纵向传播常数为 $k_c \approx \sqrt{\frac{\omega\sigma\mu_0}{2}}(1-j)$ ，自由空间波阻抗为 $Z_a=377\Omega$ ，由

5.4.12 式，铜的波阻抗为 $Z_m=R(1+j)=0.583379 \times 10^{-2} (1+j) \Omega$ 。

反射系数为： $\Gamma = \frac{Z_m - Z_a}{Z_m + Z_a} = -0.999969 + 0.00003094753j$

铜板吸收的功率的百分比为： $\eta = 1 - |\Gamma|^2 = 0.0062\%$

5.14

(1) $z=5\text{m}$ 时， $k_2 z = \frac{\pi}{4}$ ， $k_2 = \frac{\pi}{20} \text{ (rad/m)}$

$$f = \frac{k_2}{2\pi\sqrt{9\varepsilon_0\mu_0}} = \frac{\frac{\pi}{20}}{6\pi \times \frac{1}{3 \times 10^8}} = 2.5 \times 10^6 \text{ Hz} = 2.5 \text{ MHz}$$

$$(2) \quad \eta_2 = \sqrt{\frac{\mu_2}{\varepsilon_2}} = \sqrt{\frac{\mu_2}{9\varepsilon_0}} = \frac{1}{3}\eta_0 = 40\pi \text{ (}\Omega\text{)}$$

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = -\frac{1}{2}$$

$$T = 1 + \Gamma = \frac{1}{2}$$

在介质区域中 $E_2 = \eta_2 H_z = 400\pi \times 10^{-j\frac{\pi}{4}}$

$z=0$ 处，透射波 $E_{t0} = 400\pi$ ，入射波 $E_{i0} = \frac{1}{T}E_{t0} = 800\pi \text{ (mV/m)}$

在空气区域中的场强是入射波与反射波的合成，以 E_1 、 H_1 表示

$$E_1 = E_{i0}(e^{-jk_1 z} + \Gamma e^{jk_1 z}) = 800\pi \left(e^{-jk_1 z} - \frac{1}{2} e^{jk_1 z} \right) \text{ (mV/m)}$$

$$H_1 = \frac{E_{i0}}{\eta_0}(e^{-jk_1 z} - \Gamma e^{jk_1 z}) = \frac{20}{3} \left(e^{-jk_1 z} + \frac{1}{2} e^{jk_1 z} \right) \text{ (mA/m)}$$

$$k_1 = \frac{2\pi}{\lambda} = \frac{\pi}{60} \text{ (rad/m)}$$

在介质 2 中 E_2 、 H_2 为

$$H_2 = 10e^{-j\frac{\pi}{20}z} \text{ (mA/m)}$$

$$E_2 = \eta_2 H_2 = 400\pi e^{-j\frac{\pi}{20}z} \text{ (mV/m)}$$

$$(3) \quad E_2(z, t) = \text{Re}(E_2 e^{j\omega t}) = 400\pi \cos\left(\omega t - \frac{\pi}{20}z\right) \text{ (mV/m)}$$

$$H_2(z, t) = \operatorname{Re}(H_2 e^{j\omega t}) = 10 \cos\left(\omega t - \frac{\pi}{20} z\right) \quad (\text{mA/m})$$

$$S(t) = 4\pi \cos^2\left(\omega t - \frac{\pi}{20} z\right) \quad (\text{mW/m}^2)$$

$$\langle s(t) \rangle = 2\pi \quad (\text{mW/m}^2)$$

$$W_e(t) = \frac{1}{2} \varepsilon E_2^2(z, t) = \frac{9\varepsilon_0}{2} \times 160\pi \cos^2\left(\omega t - \frac{\pi}{20} z\right) = 720\pi\varepsilon_0 \cos^2\left(\omega t - \frac{\pi}{20} z\right) \quad (\text{mJ/m}^3)$$

$$W_m(t) = \frac{1}{2} \mu_0 H_2^2(z, t) = \frac{\mu_0}{2} \times 0.1 \cos^2\left(\omega t - \frac{\pi}{20} z\right) = 0.05\mu_0 \cos^2\left(\omega t - \frac{\pi}{20} z\right) \quad (\text{mJ/m}^3)$$

5.15

由图可知, $\rho = 1.5 / 0.5 = 3$

$$\text{则, } |\Gamma| = \frac{\rho - 1}{\rho + 1} = 0.5$$

由图可知, $\psi(0) = -\pi$

所以, $\Gamma = -0.5$

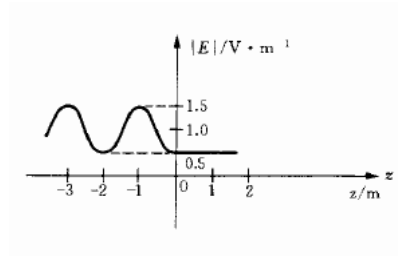
垂直投射时, $k_z = k$

$$\text{则, } \Gamma = \frac{k_1 - k_2}{k_1 + k_2} = \frac{k_0 - \sqrt{\varepsilon_{r2}} k_0}{k_0 + \sqrt{\varepsilon_{r2}} k_0} = -0.5$$

所以, $\varepsilon_{r2} = 9$

由图可知, $\lambda = 4\text{m}$

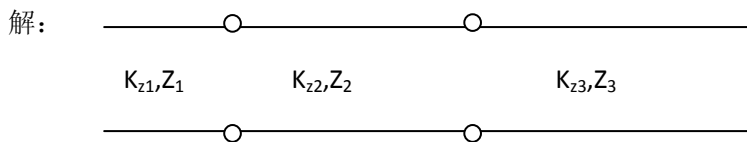
$$f = \frac{3 \times 10^8}{4} = 7.5 \times 10^7 \text{ Hz} = 75 \text{ MHz}$$



5.16 在介电系数分别为 ε_1 与 ε_3 的介质中间放置一块厚度为 d 的介质板, 其介电常数为 ε_2 , 三种介质的磁导率均为 μ_0 , 若均匀平面波从介质 1 以 $\theta^i = 0^\circ$ 垂直投射到介质板上, 试证明

当 $\varepsilon_2 = \sqrt{\varepsilon_1 \varepsilon_3}$, 且 $d = \frac{\lambda_0}{4\sqrt{\varepsilon_{r2}}}$ 时, 没有反射。

如果 $\theta^i \neq 0^\circ$, 导出没有反射时的 d 的表达式。



每一层介质可等效为传输线, 如果均匀平面波从介质 1 以 $\theta^i = 0^\circ$ 垂直投射到介质板上, 对 TE 波, 传输线的特征参数为

$$k_{z1} = \omega \sqrt{\mu_0 \varepsilon_1} = \sqrt{\varepsilon_{r1}} k_0, k_0 = \omega \sqrt{\mu_0 \varepsilon_0}, Z_1 = \omega \mu_0 / k_{z1} = \eta_0 / \sqrt{\varepsilon_{r1}}, \eta_0 = \sqrt{\mu_0 / \varepsilon_0}$$

$$k_{z2} = \omega \sqrt{\mu_0 \varepsilon_2} = \sqrt{\varepsilon_{r2}} k_0, Z_2 = \omega \mu_0 / k_{z2} = \eta_0 / \sqrt{\varepsilon_{r2}},$$

$$k_{z3} = \omega \sqrt{\mu_0 \varepsilon_3} = \sqrt{\varepsilon_{r3}} k_0, Z_3 = \omega \mu_0 / k_{z3} = \eta_0 / \sqrt{\varepsilon_{r3}},$$

当 $d = \frac{\lambda_0}{4\sqrt{\epsilon_{r2}}}$, $k_{z2}d = \pi/2$, 即介质板相当于 $\lambda/4$ 传输线, 当 $Z_2^2 = Z_1 Z_3$ 时, 传输线匹配,

即没有反射, 把波阻抗公式代入即可得 $\epsilon_2 = \sqrt{\epsilon_1 \epsilon_3}$, 所以得证。

当 $\theta^i \neq 0^\circ$, $k_{z1} = \sqrt{\epsilon_{r1}} k_0 \cos \theta$, $k_0 = \omega \sqrt{\mu_0 \epsilon_0}$, $Z_1 = \omega \mu_0 / k_{z1}$

$k_{z2} = k_0 \sqrt{\epsilon_{r2} - \epsilon_{r1} \sin^2 \theta}$, $Z_2 = \omega \mu_0 / k_{z2}$, $k_{z3} = k_0 \sqrt{\epsilon_{r3} - \epsilon_{r1} \sin^2 \theta}$, $Z_3 = \omega \mu_0 / k_{z3}$

若要求没有反射, 则 $Z_{in} = \frac{Z_3 + jZ_2 \operatorname{tg} k_{z2} d}{Z_2 + jZ_3 \operatorname{tg} k_{z2} d} = Z_1$, 此即为无反射时 d 所要满足的方程。

5.17

由 5.4.10 式, 铜的纵向传播常数为 $k_c \approx \sqrt{\frac{\omega \sigma \mu_0}{2}} (1-j)$,

由 5.4.12 式, 铜的波阻抗为 $Z_m = \sqrt{\frac{\omega \mu_0}{2\sigma}} (1+j)$

由传输线模型, 设铜的透射系数为 T , 则: $\Gamma_1(d) = 1+T$, $T = \frac{-2Z_m}{Z_0 + Z_m}$

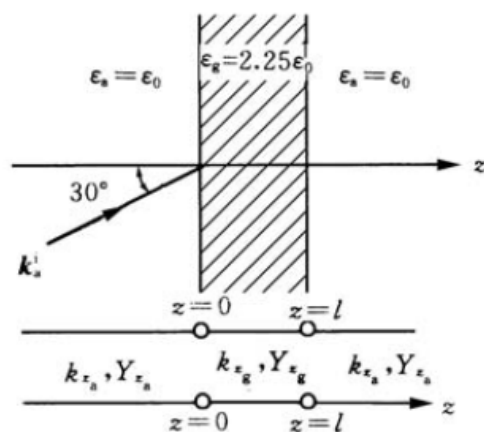
而 $\Gamma_1(0) = \Gamma_1(d) e^{-2jk_m d}$

在铜的前表面上呈现的阻抗为: $Z_{in} = \frac{1+\Gamma_1(0)}{1-\Gamma_1(0)} Z_m$

那么在铜前表面的空气中的反射系数为: $\Gamma_0 = \frac{Z_{in} - Z_0}{Z_{in} + Z_0}$

可以得到: $\frac{1+\Gamma_0}{T} = \frac{-(Z_0 + Z_m) [Z_0 + Z_m + (Z_0 - Z_m) e^{-2jkd}]}{Z_m [Z_0 + Z_m + (Z_0 - Z_m) e^{-2jkd}] + Z_0 [Z_0 + Z_m - (Z_0 - Z_m) e^{-2jkd}]}$

5.18



解题方法与步骤同教材第 246 页第五章例 5-7。

5.19 如果作增透膜，选择每一层介电系数、厚度使 $\Gamma = \frac{Z_{in} - Z_1}{Z_{in} + Z_1} \rightarrow 0$

如果作全反射膜使 $\Gamma = \frac{Z_{in} - Z_1}{Z_{in} + Z_1} \rightarrow 1$